Module 8: Network Access Basic routing and Advance routing concept, switching concept-

- **Beginner Question**

1. Explain Switch

Ans:

- It is a network communication device.
- It is a unicast devices.
- Switch are create one broadcast domain and multiple collision domain

There are different types switch:

- Layer2(datalink layer)Switch,
- Layer3 (network layer)switch,
- Access layer switch(Edge switch),
- Distribution Layer Switch,
- Core layer switch.

Function of switch are:

- Uses Mac address,
- Works at OSI layer2,
- Reduces network traffic.
- Improves speed and security,

2. Explain Switch Boot Sequence

Ans:

Switch Boot Sequence :

Switch ON,

Bios(check Hardware),

POST

Flash

NVME

Dynamic Ram(30 to 160 seconds time for booting)

Operation Mode
Learning ,
Blocking ,
Listening ,
Learning ,
Forwarding ,

Communication Mode:
Simplex
Half duplex
Full duplex

3. Explain Three Methods to access Switch Command Line Interface

Ans:

- Console Access(Local Access)
  Direct physical connection to the switch using a console cable
  You connect your PC to the switch's console port**.**

- Telnet Access(Remote Access)
  Connects to switch over a network using Telnet Protocol.
  Requires the switch to have:
    An IP Address,
    VTY Line Configuration
  Not secure (data sent in plain text)
- SSH Access (Secure Remote Access)
  Similar to telnet but uses encryption
  Requires:
    Ip address configured
    Domain name
    RSA keys generated
    Username , password.
  Protects login credentials and data
  Secure remote management.

4. Explain and Configuring the Cisco Internet Operating System

Ans:

Cisco Internet Operating System is the operating system used in Cisco networking devices to control hardware , manage network traffic , and to provide a CLI interface for configuration.

User Mode :

Eg: Switch>

Limited Access(Only basic command)

Privileged Mode:

Eg: Switch#

Access using Swtich>Enable

Full access to show and debug commands

Global Configuration Mode:

Eg: Switch(config)#

Enter using Switch#Configure terminal or conf t

Used to change device configuration

Interface Configuration Mode

Eg: Switch(config-if)

Configure specific interfaces/ports.

5. Explain Switch Port

Ans:

It is a physical interface on a network switch where devices like pc are connected.

A switch port is a connection point on a switch.

It is used to send and receive data frames.

It controls how data flows in the network.

Types of ports : Access Port and Trunk Port

Port States: Up(active) , Down(not connected) , Administratively down(Manually shut using command)

4-R1, R2, R3, and R4 have their Fast Ethernet 0/0 interfaces attached to the same VLAN. A network engineer has typed a configuration for each router by using a word processor.
He will later copy and paste the configuration into the routers. Examine the following exhibit, which lists configuration for the four routers, as typed by the network engineer. Assuming that all four routers can ping each other's LAN IP addresses after the configuration has been applied, choose the routers that will be able to form a neighbor relationship with the other routers on the LAN. (You can assume that, if not shown in the exhibit, all other related parameters are still set to their defaults.) (Choose two)

```
Configuration c
!
interface faste
   ip address 1
!
router ospf 0
   network 10.6
```

```
Configuration c
!
interface faste
   ip address 1
!
router ospf 1
   network 10.6
```

```
Configuration c
!
interface faste
   ip address 1
!
router ospf 3
   network 10.1
```

```
Configuration c
!
interface faste
   ip address 1
!
router ospf 2
   network 10.6
```

A. R1

B. R2

C. R3

D. R4

E. None of the routers will exchange routing information.

Ans:

3- enable secret [password] is hashed using the_____algorithm.

A. MD5

B. AH

C. PSK

D. ESP

E. WPA2

Ans:MD5

4- An engineer connects to Router R1 and issues a show ip ospf neighbor command. The status of neighbor 2.2.2.2 lists FULL/BDR. What does the BDR mean?

A. R1 is an Area Border Router.

B. R1 is a backup designated router.

C. Router 2.2.2.2 is an Area Border Router.

D. Router 2.2.2.2 is a backup designated router.

Ans:Router 2.2.2.2 is a backup designated router

5- Which command is used to view the neighbor discovery table on a PC?

A. show ipv6 neighbor

B. show ipv6 neighbors

C. netsh interface ipv6 show neighbor

D. netsh interface ipv6 show neighbors

E. ***Ans:*** ***netsh interface ipv6 show neighbor***

6- What type of variable is being shown? Routers = [R1,R2,R3]

A. List

B. Dictionary

C. Simple

D. Unsigned integers

Ans:List

7- Identify the fields in an IPv4 header. (Choose three)

A. Host component

B. Time to Live

C. Source address

D. Destination address

E. Network address

AnsTime to Live , Source Address , Destination Address